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Q1.

b. Queue

Q3.

False. A primary key cannot contain NULL values because it must uniquely identify every row.

Q2. Random Access Memory

Q5.

b. Quick sort

Q6. Time complexity is a measure of the amount of time an algorithm takes to run, expressed as a function of the size of its input. Big-O notation is used because it describes the upper bound on the growth rate, which lets us compare algorithms without depending on the hardware.

Q8. Normalization is the process of organising the data in a database so that redundancy is reduced and update anomalies are avoided. Two normal forms are First Normal form (1NF) and Second Normal form (2NF)

Arka

Q10. Deadlock is a situation in which two or more processes are each waiting for a resource that is held by another process, so none of them can ~~so~~ ever proceed. One necessary condition is mutual exclusion.

Q9. TCP is connection oriented whereas UDP is connectionless. TCP guarantees delivery and preserves ordering, while UDP does neither. TCP is slower because of acknowledgements, so UDP is preferred for live streaming.

Q11.

a. A hash table stores key-value pairs. A hash function converts the key into an index of an underlying array and the value is placed at that index, which gives avg. constant-time lookup.

Q11. b. A collision occurs when 2 different keys are hashed to the same index. One resolution technique is seperate chaining, in which every index holds a linked list of all the entries that hash to it.

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Q12.

Virtual memory is a memory management technique in which the system gives the illusion of very large main memory by using secondary storage as an extension of it. In paging, the logical address space of a process is divided into fixed size blocks called pages, and physical memory is divided into frames of the same size. A page table maps every page number to a frame number. When a process refers to a page that is not present in main memory a page fault is raised and the

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operating systems loads that page from the disk into a free frame. Two advantages are that a process larger than the available physical memory can still be executed, and the degree of multiprogramming is increased.

Q13. The TCP/IP model is a 4 layer networking model made up of the Link layer, the Internet layer, the Transport layer and the Application layer. Compared with the OSI model it has for layers instead of seven. The OSI session, Presentation and Application layers are merged into the single TCP/IP Application layer, the OSI Physical layer and Data Link layers are merged into the Link layer.

Q14.

b. 45 divided by 2 gives 22 remainder 1. 22 divided by 2 gives 11 remainder 0. 11 divided by 2 gives 5 remainder. 5 divided by 2 gives 2 remainder 1. 2 divided by 2 gives 1 remainder 0. 1 divided by 2 gives 0 remainder 1. Reading the remainders upwards. 45 binary is 101101.

Q14 a. Recursion is a technique in which a function calls itself in order to solve a smaller instance of the same problem. For example factorial(n) is defined as n multiplied by factorial(n-1), with factorial(0) equal to 1.

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Q4. Hyper Text Transfer Protocol

Q5. Big Data refers to data sets that are so large or so complex that traditional data processing tools cannot handle them.

Rough work

101101

1024 512 256 128 64 32 16 8 4 2 1